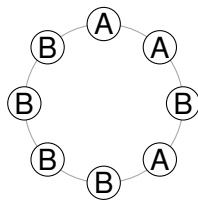
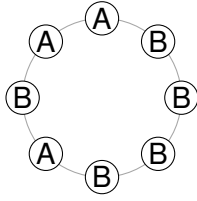


Counter kinds keep their names. Turning a ring may match it to another ring. For Problem 1, test flips with a tracing copy while the original stays still.

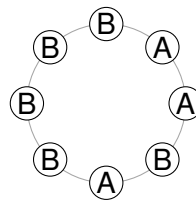
**Problem 1:** Build and compare these six rings. Group the ring labels when only turns are allowed. Then group them when turns and flips are allowed.



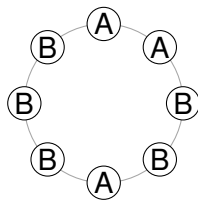
a



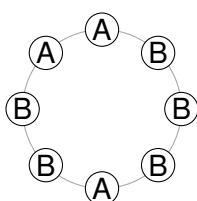
b



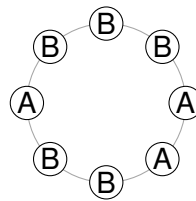
c



d



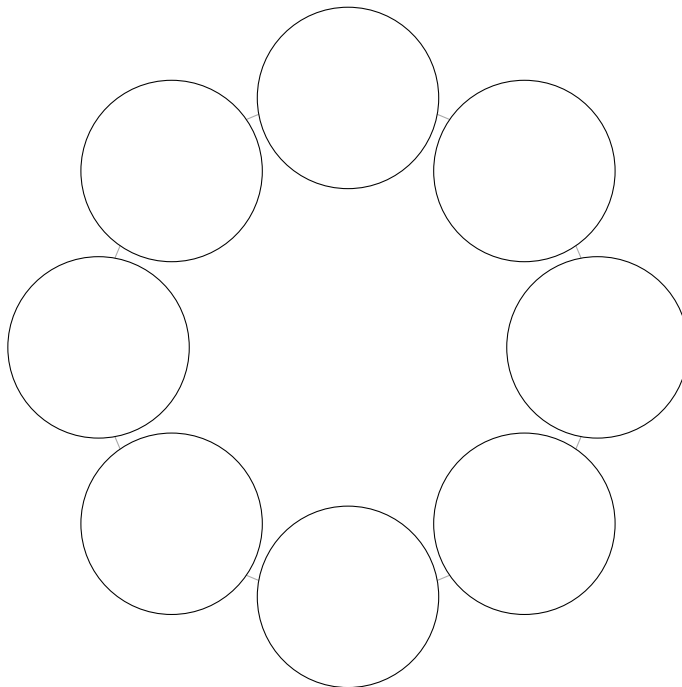
e



f

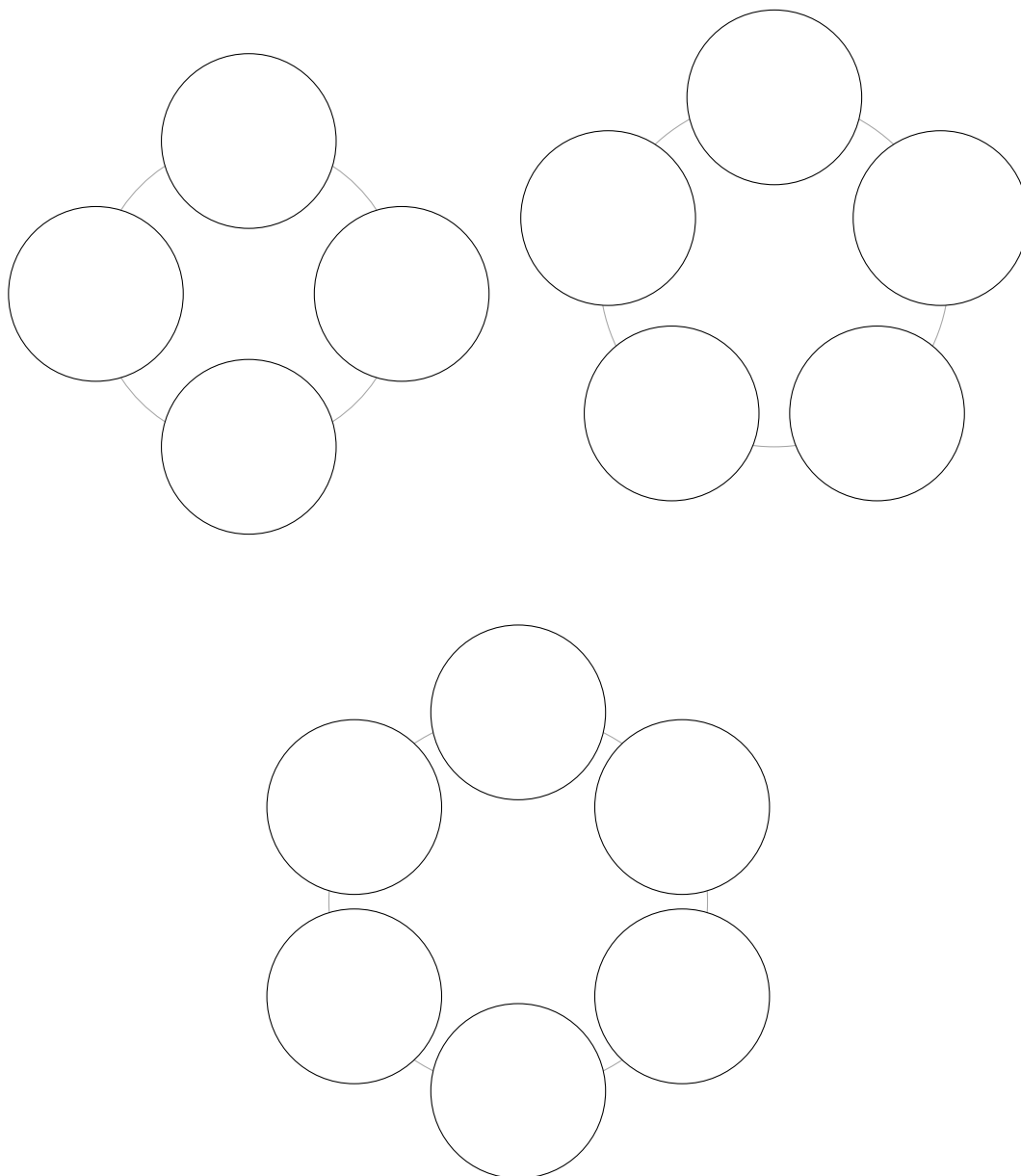
Only turns: \_\_\_\_\_

Turns and flips: \_\_\_\_\_

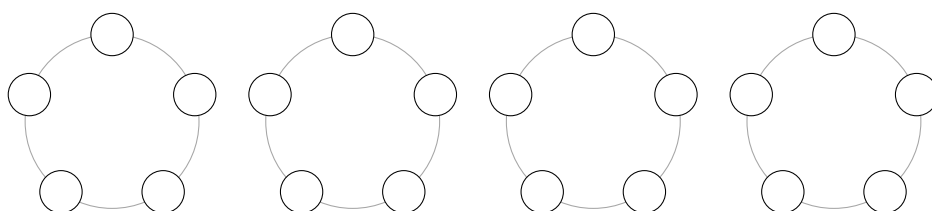
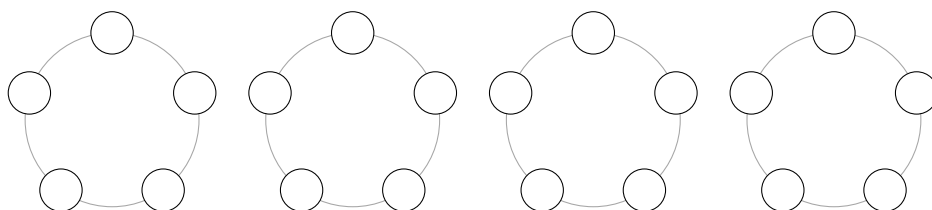
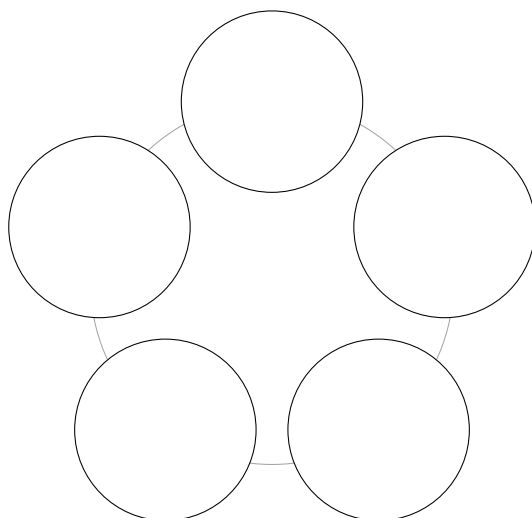


For the remaining problems, rings stay face up. Only turns count as matches.

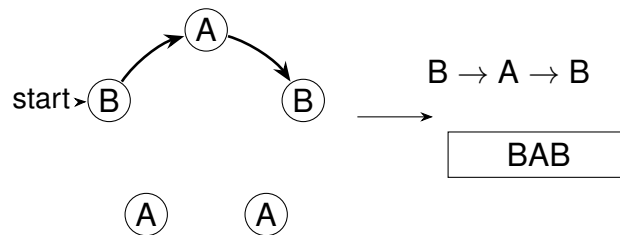
**Problem 2:** Make each ring using only A and B, with different kinds at every neighboring pair, including the closing pair. Which rings can be made? Explain any impossibility.



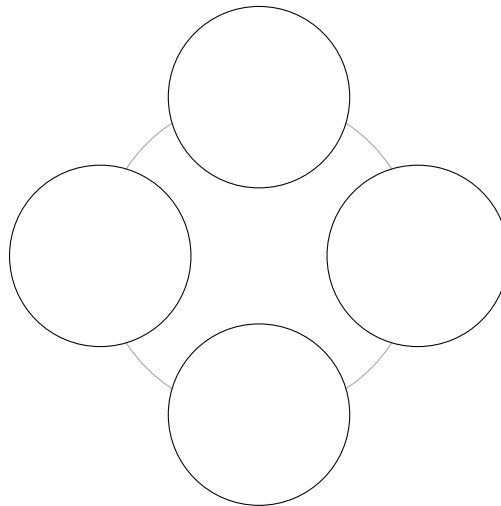
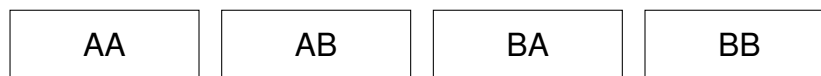
**Problem 3:** Use A, B, and C to make five-bead rings with different kinds at every neighboring pair. Find every ring. Rings that match by turning count as one; flips do not count as matches.



A window reads consecutive beads clockwise. Start at every bead, including windows that cross the closing gap. A removable viewer marks which beads to read. For each test, use a fresh complete deck. Your partner removes one matching card at every start. If it is already gone, the ring fails.



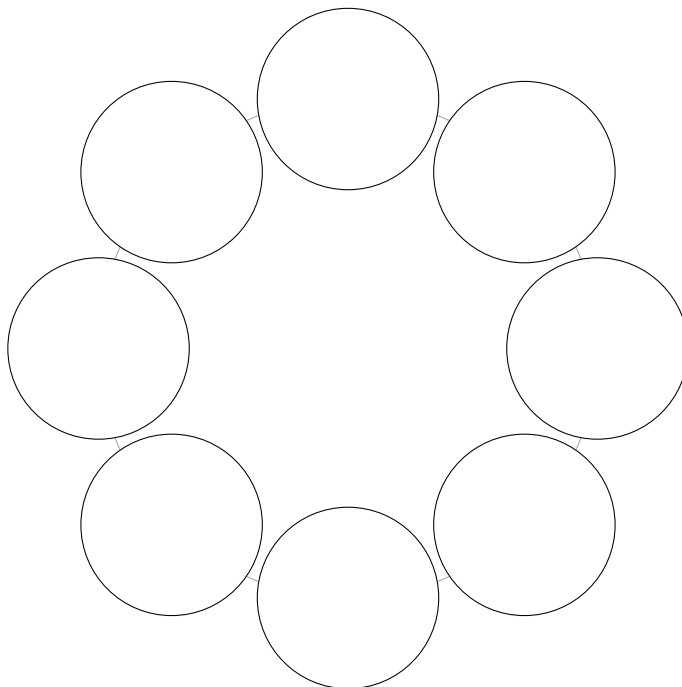
**Problem 4:** Make the shortest A/B ring that contains each two-letter card exactly once as a window.



**Problem 5:** Why can no shorter ring contain all four cards?

**Problem 6:** Make the shortest A/B ring that contains each three-letter card exactly once as a clockwise window.

|     |     |     |     |
|-----|-----|-----|-----|
| AAA | AAB | ABA | ABB |
| BAA | BAB | BBA | BBB |



**Problem 7:** Find every shortest ring for the eight cards. Rings that match by turning count as one. How are their flipped copies related to your collection?

